Ex - 02	IPY Widgets - Electricity Bill Management System

Aim

To develop an **Electricity Bill Management System** using Python and the ipywidgets library that collects customer details, calculates electricity charges based on meter readings, applies tax and discount, and displays the final electricity bill interactively.

Problem Statement:

To develop an interactive Electricity Bill Management System using Python that collects customer details, calculates electricity charges based on meter readings and customer type, applies taxes and discounts, and generates a detailed bill summary with payment information.

Procedure

1. Import the required libraries such as ipywidgets, IPython.display, datetime, and random.
2. Create input widgets to collect customer details, including name, consumer number, address, and mobile number.
3. Create a dropdown widget to select the customer type as Domestic or Commercial.
4. Create input widgets for previous and current electricity meter readings.
5. Add a slider to set the tax percentage and a checkbox to apply the green discount.
6. Create a dropdown widget to select the payment mode.
7. Define a function to validate meter readings and calculate units consumed.
8. Calculate energy charges, tax amount, discount, and the total bill.
9. Display the bill number, date, customer details, charges, and payment status.
10. Create three interactive tabs for customer details, bill details, and bill summary, and display the interface.

```
import ipywidgets as widgets
from IPython.display import display, clear_output
from datetime import date
import random

# ====================================
# CUSTOMER DETAILS
# ====================================

name = widgets.Text(
    description="Name:",
    placeholder="Enter Name"
)

consumer = widgets.IntText(
    description="Consumer:"
)

address = widgets.Textarea(
    description="Address:",
    placeholder="Enter Address"
)

mobile = widgets.Text(
    description="Mobile:",
    placeholder="Enter Mobile Number"
)

ctype = widgets.Dropdown(
    options=["Domestic", "Commercial"],
    description="Type:"
)

tab1 = widgets.VBox([
    widgets.HTML("<h3>Customer Details</h3>"),
    name,
    consumer,
    address,
    mobile,
    ctype
])

# ====================================
# BILL DETAILS
# ====================================

previous = widgets.IntText(
    description="Previous:"
)

current = widgets.IntText(
    description="Current:"
)

tax = widgets.FloatSlider(
    value=5,
    min=0,
    max=20,
    step=1,
    description="Tax %"
)

green = widgets.Checkbox(
    value=False,
    description="Green Discount ₹50"
)

payment = widgets.Dropdown(
    options=["Cash", "UPI", "Card", "Net Banking"],
    description="Payment:"
)

button = widgets.Button(
    description="Calculate Bill",
    button_style="success",
    icon="calculator"
)

tab2 = widgets.VBox([
    widgets.HTML("<h3>Bill Details</h3>"),
```

```
    current,
    tax,
    green,
    payment,
    button
])

# =====================================
# BILL SUMMARY
# =====================================

output = widgets.Output()

tab3 = widgets.VBox([
    widgets.HTML("<h3>Electricity Bill Summary</h3>"),
    output
])

# =====================================
# BILL CALCULATION FUNCTION
# =====================================

def bill_calculate(b):

    with output:

        clear_output()

        # Validation
        if current.value < previous.value:
            print("❌ Error: Current reading must be greater than previous reading.")
            return

        units = current.value - previous.value

        # Rate Selection
        if ctype.value == "Domestic":
            rate = 6
        else:
            rate = 8

        energy = units * rate
        tax_amount = energy * tax.value / 100
        discount = 50 if green.value else 0
        total = energy + tax_amount - discount

        bill_no = random.randint(100000, 999999)

        print("=" * 50)
        print("            ELECTRICITY BILL")
        print("=" * 50)

        print(f"Bill Number      : {bill_no}")
        print(f"Bill Date        : {date.today()}")

        print("\nCustomer Details")
        print("-" * 50)
        print(f"Name             : {name.value}")
        print(f"Consumer No      : {consumer.value}")
        print(f"Mobile           : {mobile.value}")
        print(f"Address          : {address.value}")
        print(f"Customer Type    : {ctype.value}")

        print("\nMeter Details")
        print("-" * 50)
        print(f"Previous Reading : {previous.value}")
        print(f"Current Reading  : {current.value}")
        print(f"Units Consumed   : {units}")

        print("\nCharges")
        print("-" * 50)
        print(f"Rate per Unit    : ₹{rate}")
        print(f"Energy Charge    : ₹{energy:.2f}")
        print(f"Tax ({tax.value}%)      : ₹{tax_amount:.2f}")
        print(f"Discount         : ₹{discount:.2f}")

        print("-" * 50)
        print(f"TOTAL BILL       : ₹{total:.2f}")
        print("-" * 50)

        print(f"Payment Mode     : {payment.value}")
        print("Status           : Pending")
```

```
# Button Click Event
button.on_click(bill_calculate)

# ======================================
# CREATE TABS
# ======================================

tabs = widgets.Tab(children=[tab1, tab2, tab3])

tabs.set_title(0, "Customer Details")
tabs.set_title(1, "Bill Details")
tabs.set_title(2, "Bill Summary")

display(tabs)
```

Customer Details | Bill Details | Bill Summary

Electricity Bill Summary

```
================================================
            ELECTRICITY BILL
================================================
Bill Number         665563
Bill Date         : 2026-10-20

Customer Details
------------------------------------------------
Name              : CHANDRU P
Consumer No         1
Mobile              842601537
Address           : 1/29,pettai street, kovilur, koilur.
Customer Type     : Domestic

Meter Details
------------------------------------------------
Previous Reading : 100
Current Reading     200
Units Consumed      100

Charges
------------------------------------------------
Rate per Unit     : ₹6
Energy Charge     : ₹600.00
Tax (5.0%)       : ₹30.00
Discount          : ₹0.00
------------------------------------------------
TOTAL BILL        : ₹630.00
------------------------------------------------
Payment Mode      : Cash
Status            : Pending
```

Visualization:

Interactive tab-based GUI for entering customer details, billing information, and displaying the electricity bill.

Insights:

Automatically calculates the total electricity bill based on units consumed, customer type, tax, and discount.

Result:

The Electricity Bill Management System was successfully developed using Python and ipywidgets. It accurately calculates and displays the electricity bill based on customer details, meter readings, tax, and applicable discounts.